

The Three Geometric Constants:

π , ϖ , and G^*

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Abstract

We identify three geometric constants—the circular constant π , the lemniscate constant ϖ , and the lattice bridge constant G^* —each defined by a canonical integral over a distinct geometric domain: the circle, the lemniscate, and the cubic lattice $\mathbb{Z}^3$. We prove that the three satisfy the triad relation $\pi = 4\varpi^2/G^{*2}$ and show that this relation is a consequence of all three constants being periods of the CM elliptic curve $E: y^2 = x^3 - x$. We argue that G^* plays a role for discrete lattice geometry analogous to the role π plays for circular geometry and ϖ plays for lemniscatic geometry: it is the canonical measure of the lattice’s self-interaction, defined independently of the other two but related to them by an exact algebraic identity. No physical interpretation is assumed.

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1 Introduction

The circle constant π has been known since antiquity. The lemniscate constant ϖ was studied by Euler in the 1750s and appears in Gauss’s mathematical diary (1796). Both are now recognized as periods of elliptic curves: π of the degenerate curve $y^2 = 1 - x^2$, and ϖ of the CM curve $E: y^2 = x^3 - x$.

We identify a third constant, $G^* = \sqrt{2} \Gamma(1/4)^2 / (2\pi) \approx 2.9587$, that completes a natural triad. While G^* has appeared implicitly in lattice combinatorics since Watson’s 1939 evaluation of the cubic lattice Green’s function, its role as a geometric constant on par with π and ϖ has not been articulated. We make this case here.

The three constants are not algebraically independent—they satisfy the exact relation $\pi = 4\varpi^2/G^{*2}$. But algebraic dependence does not disqualify a constant from being geometrically fundamental. The constant π itself satisfies $\pi = \varpi/M$ where M is the arithmetic-geometric mean of 1 and $\sqrt{2}$; this relation, discovered by Gauss, does not diminish π ’s status. What makes a constant “fundamental” is not algebraic irreducibility but *geometric irreplaceability*: each constant is the natural measure of a distinct geometric domain.

2 Three Constants, Three Integrals

Definition 2.1 (The Circular Constant).

$$\pi = 4 \int_0^1 \frac{dx}{1+x^2} = 3.14159265 \dots \quad (1)$$

The half-perimeter of the unit circle. Equivalently, the area of the unit disk, or the period of $\sin$ and $\cos$. The integrand $1/(1+x^2)$ is *rational* in x .

Definition 2.2 (The Lemniscate Constant).

$$\varpi = 2 \int_0^1 \frac{dx}{\sqrt{1-x^4}} = 2.62206 \dots \quad (2)$$

The half-perimeter of the unit lemniscate $r^2 = \cos 2\theta$. The integrand $(1-x^4)^{-1/2}$ involves a *quartic* under the radical—the simplest algebraic integrand that produces a genuinely new transcendental.

Definition 2.3 (The Lattice Bridge Constant).

$$G^{*2} = \frac{2\pi}{(2\pi)^3} \int_{[-\pi, \pi]^3} \frac{d^3 k}{3 - \cos k_1 - \cos k_2 - \cos k_3} = 2\pi W_3 \quad (3)$$

where W_3 is Watson's integral for the simple cubic lattice $\mathbb{Z}^3$. Equivalently, $G^* = \sqrt{2} \Gamma(1/4)^2 / (2\pi) \approx 2.9587$. The integrand is a *trigonometric sum*—the inverse of the lattice Laplacian in momentum space.

Remark 2.4 (Dimensional progression). The three integrals live in progressively higher dimensions:

- π : a 1-dimensional integral over $[0, 1]$, defining a property of a curve in $\mathbb{R}^2$.
- ϖ : a 1-dimensional integral over $[0, 1]$, but involving a quartic (degree 4) algebraic curve.
- G^* : a 3-dimensional integral over the Brillouin zone $[-\pi, \pi]^3$ of $\mathbb{Z}^3$.

The algebraic complexity increases at each step: rational $\rightarrow$ quartic radical $\rightarrow$ trigonometric lattice sum.

3 The Triad Relation

Theorem 3.1 (The Triad). *The three constants satisfy:*

$$\boxed{\pi = \frac{4\varpi^2}{G^{*2}}} \quad (4)$$

Equivalently: $G^{*2}/\varpi^2 = 4/\pi$, or $\varpi/G^* = \sqrt{\pi}/2$.

Proof. From the definitions: $\varpi = \Gamma(1/4)^2 / (2\sqrt{2\pi})$ and $G^* = \sqrt{2} \Gamma(1/4)^2 / (2\pi)$. Then:

$$\frac{4\varpi^2}{G^{*2}} = \frac{4 \cdot \Gamma(1/4)^4 / (8\pi)}{2 \Gamma(1/4)^4 / (4\pi^2)} = \frac{4/(8\pi)}{2/(4\pi^2)} = \frac{4}{8\pi} \cdot \frac{4\pi^2}{2} = \frac{16\pi^2}{16\pi} = \pi. \quad \square$$

Remark 3.2 (Historical parallel). Gauss discovered the relation $\pi = \varpi/M$ where $M = \text{agm}(1, \sqrt{2})$ is the arithmetic-geometric mean. The triad relation (4) extends this: $G^* = 2\varpi/\sqrt{\pi} = 2\varpi \cdot M/\sqrt{\varpi M} = 2\sqrt{\varpi M}$. Thus G^* is twice the geometric mean of the lemniscate constant and the AGM. The three relations form a triangle:

$$\pi = \varpi/M \quad (\text{Gauss}) \quad (5)$$

$$G^* = 2\varpi/\sqrt{\pi} \quad (\text{bridge}) \quad (6)$$

$$\pi = 4\varpi^2/G^{*2} \quad (\text{triad}) \quad (7)$$

Any two of these imply the third.

4 Three Geometric Operations

Each constant is the canonical measure of a distinct geometric operation.

4.1 π and closure

The circle is the unique curve that *closes*: it returns to its starting point after traversing its full length. The constant π measures this closure—it is half the length needed to close. Every periodic function with period T involves π through $\omega = 2\pi/T$. Closure is a topological property: the circle is contractible to a point but not to a segment.

4.2 ϖ and self-crossing

The lemniscate is the simplest algebraic curve that *crosses itself*. Its node at the origin is where the curve meets itself, creating two lobes. The constant ϖ measures this self-crossing geometry. While the circle encloses space, the lemniscate *partitions* it: the self-crossing divides the curve into two topologically distinct regions. The quartic exponent (x^4 in the integrand) is the minimal degree that allows algebraic self-crossing at a smooth node.

4.3 G^* and self-energy

The Watson integral W_3 measures the cubic lattice's *self-energy*: the probability that a random walk on $\mathbb{Z}^3$ returns to its origin, or equivalently, the diagonal element of the lattice Green's function. The constant $G^* = \sqrt{2\pi W_3}$ is the natural rescaling that converts this self-energy into a coupling amplitude. While π measures a curve's interaction with *space* (closure) and ϖ measures a curve's interaction with *itself* (crossing), G^* measures a lattice's interaction with itself (self-energy). This is a third geometric operation, irreducible to the first two.

Proposition 4.1 (The three operations are distinct). *No two of the three operations (closure, self-crossing, self-energy) are equivalent:*

1. *Closure does not require self-crossing (the circle does not cross itself).*
2. *Self-crossing does not require closure (an infinite lemniscatic spiral crosses itself without closing).*
3. *Self-energy does not require either (the lattice Green's function is defined for open lattices).*

5 A Common Ancestor

Despite their distinct geometric roles, the three constants share a common algebraic origin.

Theorem 5.1 (Common CM Curve). *All three constants are periods of the CM elliptic curve $E: y^2 = x^3 - x$ with $j(E) = 1728$, $\text{Aut}(E) \cong \mathbb{Z}/4\mathbb{Z}$, and endomorphism ring $\mathbb{Z}[i]$:*

1. ϖ is the real period of E .
2. π is related to ϖ through the AGM, which computes the period ratio of E .
3. G^* is related to ϖ through W_3 , which evaluates to a product of the periods of E via Watson's contour integration [1].

The triad relation $\pi = 4\varpi^2/G^{*2}$ is therefore not a coincidence but a consequence of all three constants being different projections of the same CM curve. The curve E is the common ancestor; the three constants are its shadows in different geometric domains.

6 The Period Descent Perspective

A Diophantine condition is proven for when the circular period π can be eliminated from the mixed Gaussian trace $\mathcal{Z}(p, q)$. The fundamental solution $(p, q) = (3, 2)$ produces $\mathcal{Z}(3, 2) = \sqrt{2}\Gamma(1/4)^2$, which is π -free and lies in the period ring of E .

Within this framework, the three constants occupy distinct positions:

- π is the *circular period*—present whenever $2p \neq 3q$.
- ϖ is the *elliptic period*—the irreducible period of E that survives descent.
- G^* is the *bridge period*—the specific combination $2\varpi/\sqrt{\pi}$ that converts between the π -dependent and π -free regimes.

The existence of exactly three period types at the $(3, 2)$ resonance is a structural property of the Chowla–Selberg formula at the self-dual point $\tau = i$.

7 Representations

We collect all known representations of the three constants, organized by source domain.

Constant	Representation	Source
π	$4\int_0^1 dx/(1+x^2)$	Leibniz–Gregory
π	$4\varpi^2/G^{*2}$	Triad relation
π	$\varpi/M \quad (M = \operatorname{agm}(1, \sqrt{2}))$	Gauss
ϖ	$2\int_0^1 dx/\sqrt{1-x^4}$	Euler
ϖ	$\Gamma(1/4)^2/(2\sqrt{2\pi})$	Beta function
ϖ	$G^*\sqrt{\pi}/2$	Bridge relation
G^*	$\sqrt{2\pi}W_3 \quad (W_3 = \text{Watson integral})$	Watson (1939)
G^*	$\sqrt{2}\Gamma(1/4)^2/(2\pi)$	Definition
G^*	$2\varpi/\sqrt{\pi}$	Bridge relation
G^*	$\sqrt{2\pi}\theta_3(e^{-\pi})^2$	Chowla–Selberg
G^*	$2\sqrt{\pi}/\operatorname{agm}(1, \sqrt{2})$	Gauss–AGM
G^*	$B(1/4, 1/4)/\sqrt{2\pi}$	Beta function
G^*	$(2\sqrt{2}/\sqrt{\pi})K(1/\sqrt{2})$	Elliptic integral

Remark 7.1. Every row is independently verifiable. The proliferation of representations for G^* reflects its position at the nexus of multiple mathematical domains—modular forms, lattice combinatorics, special functions, and elliptic integrals. This is analogous to π ’s hundreds of known representations, which reflect its centrality in mathematics.

8 Numerical Values

For reference, to 25 significant figures:

Constant	Value
π	3.141 592 653 589 793 238 462 643 ...
ϖ	2.622 057 554 292 119 810 464 840 ...
G^*	2.958 675 119 188 638 892 310 821 ...
$\Gamma(1/4)$	3.625 609 908 221 908 311 930 685 ...
ϖ/G^*	$0.886\,226\,925\,452\,758\dots = \sqrt{\pi}/2$
G^*/ϖ	$1.128\,379\,167\,095\,513\dots = 2/\sqrt{\pi}$
$4\varpi^2/G^{*2}$	$3.141\,592\,653\,589\,793\dots = \pi$

Remark 8.1 (All three from one function). The triad reduces to a single function at a single point. Since $\pi = \Gamma(1/2)^2$:

$$\varpi = \Gamma(1/4)^2 / (2\sqrt{2}\Gamma(1/2)) \quad (8)$$

$$G^* = \sqrt{2}\Gamma(1/4)^2 / (2\Gamma(1/2)^2) \quad (9)$$

$$\pi = \Gamma(1/2)^2 \quad (10)$$

The three “geometric constants” are three algebraic projections of one transcendental: $\Gamma(1/4)$. The triad relation $\pi = 4\varpi^2/G^{*2}$ is the algebraic identity connecting these projections. The Gamma function itself is forced by the Bohr–Mollerup theorem (the unique log-convex factorial extension), and the argument $1/4$ is selected by the $\mathbb{Z}_4$ symmetry of the cubic lattice.

9 Discussion: The Case for G^*

Why should G^* be treated as a fundamental geometric constant rather than “merely” $2\varpi/\sqrt{\pi}$?

1. **Independent definition.** G^* can be defined purely from the cubic lattice $\mathbb{Z}^3$ via Watson’s integral, with no reference to π or ϖ . The identity $G^{*2} = 2\pi W_3$ is a *theorem* connecting two independently defined quantities, not a definition of one in terms of the other.
2. **Distinct geometric role.** π measures closure, ϖ measures self-crossing, G^* measures self-energy. These are three irreducible geometric operations (Proposition 4.1).
3. **Algebraic dependence is not disqualifying.** π satisfies $\pi = \varpi/M$ (Gauss’s AGM relation); this does not make π “merely a rescaled ϖ .” Algebraic relations between constants reflect shared structure, not redundancy.

4. **Notational economy.** The constant G^* appears in the Watson identity, the master quadratic, the bridge relation, the Jacobi theta evaluation, the Beta function identity, and the elliptic integral at the lemniscatic modulus. Writing G^* rather than $2\varpi/\sqrt{\pi}$ each time is not merely convenient—it reflects the fact that G^* , not ϖ , is the natural coefficient in these contexts.
5. **The triad is complete.** π , ϖ , and G^* satisfy exactly one algebraic relation (eq. 4), leaving two algebraically independent constants. This is the maximum number of independent constants derivable from the single transcendental $\Gamma(1/4)$, given that π and $\Gamma(1/4)$ are algebraically independent (Chudnovsky, 1984). The triad is therefore *complete*: no fourth constant of the same type exists.

10 Summary

1. Three geometric constants— π (circle), ϖ (lemniscate), G^* (lattice)—each arise from a canonical integral over a distinct geometric domain.
2. They satisfy the triad relation $\pi = 4\varpi^2/G^{*2}$, which follows from their shared origin as periods of the CM curve $E: y^2 = x^3 - x$.
3. Each constant measures a distinct geometric operation: closure (π), self-crossing (ϖ), self-energy (G^*). These operations are irreducible to one another.
4. The triad is complete: given the algebraic independence of π and $\Gamma(1/4)$, exactly two independent constants can be formed from $\Gamma(1/4)$, and the three constants $\{\pi, \varpi, G^*\}$ with one relation exhaust this count.
5. No physical interpretation is assumed. The case for G^* rests on lattice geometry, the Watson integral, and the theory of CM elliptic curves.

References

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